0. Introduction

1. Nature of individual life: Death process
2. Nature of individual life: Childbearing

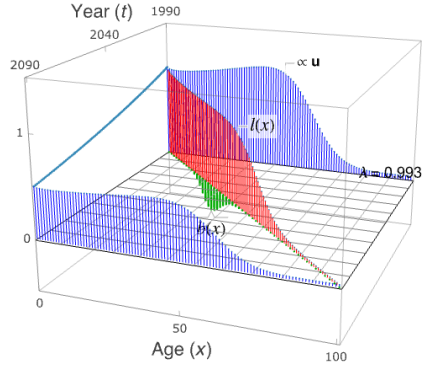


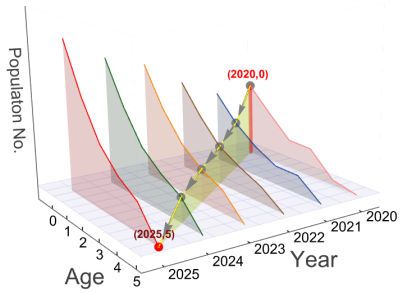
3. Characteristics of population that arise from the nature of individual's life



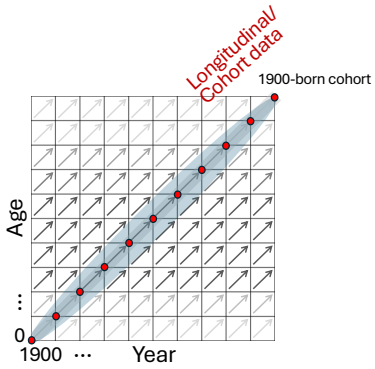
4. Traits of individuals defined in the population

View of an Ideal Population

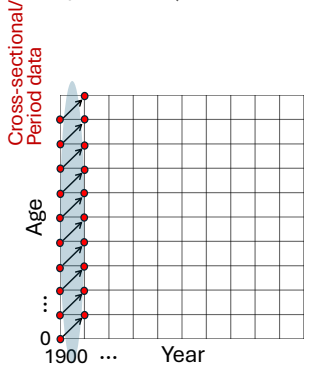



Cohort Age Specific data↗ : $d_x$, $q_x$, or $s_x$

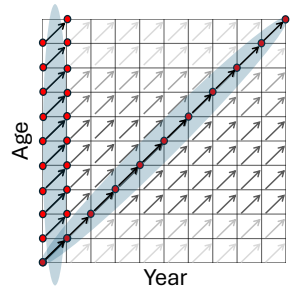


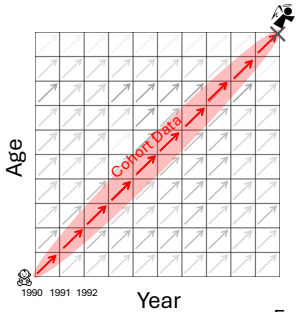

Period Age Specific data↗ : $d_{x,1900}$, $q_{x,1900}$, or $s_{x,1900}$






How should we think these two different sets of data?






Make the Cohort Survivals

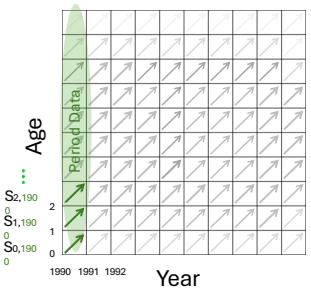
$l_{1,\text{cohort}} = s_{0,1900}$

$l_{2,\text{cohort}} = s_{0,1900}s_{1,1901}$

$l_{3,\text{cohort}} = s_{0,1900}s_{1,1901}s_{2,1902}$

⋮

Expected lifetime : $\sum l_{i,\text{ cohort}}$




Make the Period Survivals

$l_{1,\text{period}} = s_{0,1900}$

$l_{2,\text{period}} = s_{0,1900}s_{1,1900}$

$l_{3,\text{cohort}} = s_{0,1900}s_{1,1901}s_{2,1902}$

⋮

Expected lifetime : $\sum l_{i,\text{ period}}$

## 2023 Period data of Japan

Survival Data

| Year | Age | qx | sx | lx | dx | ex |
|---|---|---|---|---|---|---|
| 2023 | 0 | 0.0018 | 0.9982 | 1 | 176 | 84.16 |
| 2023 | 1 | 0.0003 | 0.9997 | 0.99824 | 27 | 83.31 |
| 2023 | 2 | 0.0002 | 0.9998 | 0.99796 | 16 | 82.33 |
| 2023 | 3 | 0.0001 | 0.9999 | 0.9978 | 13 | 81.35 |
| 2023 | 4 | 0.0001 | 0.9999 | 0.99767 | 11 | 80.36 |
| 2023 | 5 | 8.00E-05 | 0.99992 | 0.99755 | 8 | 79.37 |
| 2023 | 6 | 8.00E-05 | 0.99992 | 0.99747 | 8 | 78.37 |
| 2023 | 7 | 7.00E-05 | 0.99993 | 0.9974 | 6 | 77.38 |
| 2023 | 8 | 7.00E-05 | 0.99993 | 0.99733 | 7 | 76.38 |
| 2023 | 9 | 6.00E-05 | 0.99994 | 0.99726 | 6 | 75.39 |
| 2023 | ... | ... | ... | ... | ... | ... |

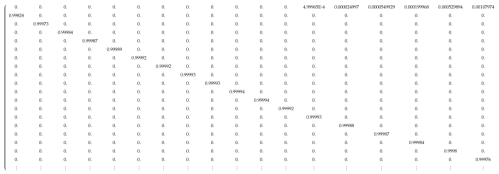
Fertility Data

| Age | bx |
|---|---|
| -12 | 0 |
| 13 | 0.00001 |
| 14 | 0.00005 |
| 15 | 0.00011 |
| 16 | 0.0004 |
| 17 | 0.00106 |
| 18 | 0.00216 |
| 19 | 0.00442 |
| 20 | 0.00769 |
| 21 | 0.01097 |
| ... | ... |

I made the Leslie Matrix based on 2023 (2022-2023) cross-sectional data (see the next slied).

Japan2023 LifeTable+Fertility.xls

L =



L is 110 by 110 dimension matrix

$\lambda_{max} = 0.983622$

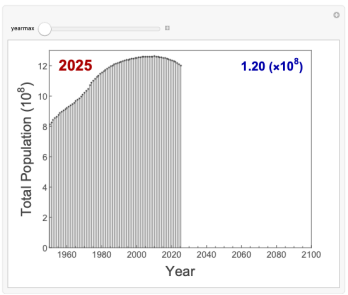
Quiz. How many years do you estimate the population will reach half its current size?

What do you think when you see that estimate?

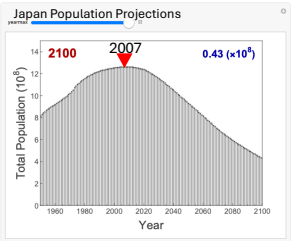
Japan Popul Projec Leslie Model.cdf

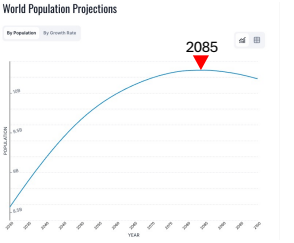
## Past, Current, and Future Population Projections

$$N_{total,t} = \lambda^t N_{total,2025}, \quad (t = 2026, \cdots)$$




Japan Leslie Mode Population.cdf




https://worldpopulationreview.com/

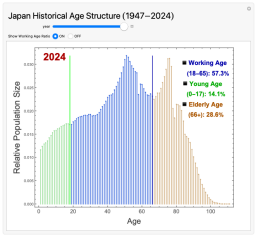
## Past, Current, and Future Population Profiles

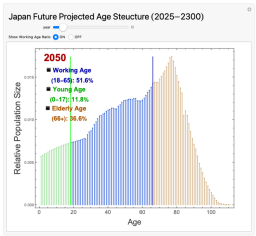
Based on 2024's demographic parameters and Age-specific Population Profile.

$$L^{t}_{2025}N_{2025}$$

Past, Current
Japan, $l_x$ (1947-

Future Population Projection

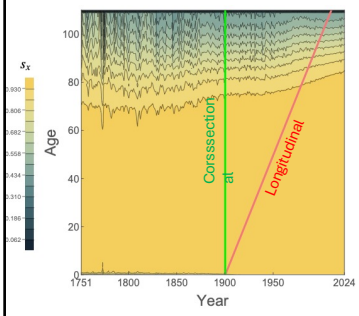



Japan Popul Projection by Leslie Model.cdf

## Historical demographic data

Sweden Age-Specific Survival Date $s_{x,year}$ (1751-2024)




Make the Cohort Survivals

$$l_{1,cohort} = s_{0,1900}$$

$$l_{2,cohort} = s_{0,1900}s_{1,1901}$$

$$l_{3,cohort} = s_{0,1900}s_{1,1901}s_{2,1902}$$

$$\vdots$$

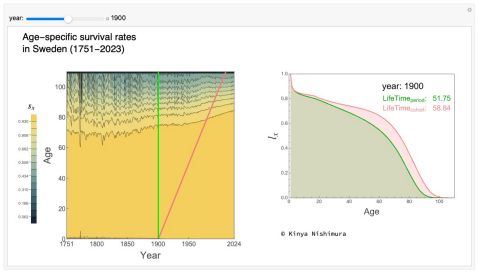
Make the Period Survivals

$$l_{1,period} = s_{0,1900}$$

$$l_{2,period} = s_{0,1900}s_{1,1900}$$

$$l_{3,cohort} = s_{0,1900}s_{1,1901}s_{2,1902}$$

$$\vdots$$

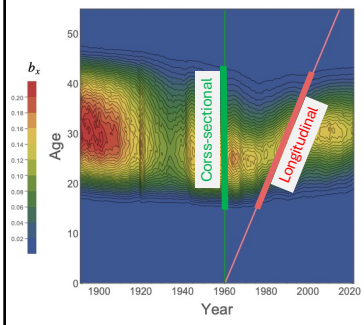



It is possible to consider why the LifeTime$_{cohort}$ is longer than the LifeTime$_{period}$, or vice versa, from several different perspectives, different academic disciplinary hierarchies.

Period_Cohort_Survival_Sweden_landscape.nb

## Historical demographic data

Sweden Age-Specific Fertility Date $b_{x,year}$ (1891-2022)



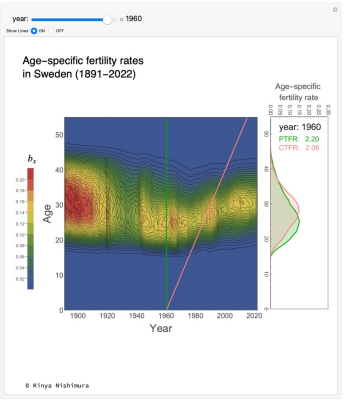

Cross-sectional/Period

$$(b_{12,1960}, b_{13,1960}, b_{14,1960}, \cdots)$$

Longitudinal/Cohort

$$(b_{12,1973}, b_{13,1974}, b_{14,1975}, \cdots)$$

PTFR.CTFR.Sweden1891-.cdf




Cross-sectional/Period

$$(b_{12,1960}, b_{13,1960}, b_{14,1960}, \cdots)$$

Longitudinal/Cohort

$$(b_{12,1973}, b_{13,1974}, b_{14,1975}, \cdots)$$

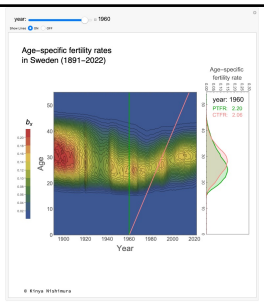
Period Total Fertility Rate

$$\mathrm{PTFR}_{year} = \sum_{age} b_{age,year}$$

Cohort Total Fertility Rate

$$\mathrm{CTFR}_{year} = \sum_{age} b_{age,year+age}$$

PTFR.CTFR.Sweden1891-.cdf



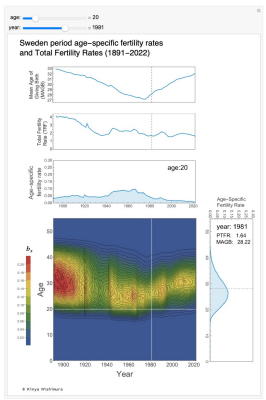

It is possible to consider why the CTFR is longer than the PTFR, or vice versa, from several different perspectives, different academic disciplinary hierarchies.

PTFR.CTFR.Sweden1891-.cdf

Historical demographic data

Sweden ASFR, $b_{x,year}$ (1891-2022)

How do you evaluate the fact presented in the graph from several different perspectives, different academic disciplinary hierarchies?



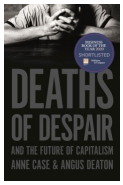

HistoricalFertilityRate.Sweden1891-2022.cdf



Analyzing Society Through the Lens of Demography

Topic 1

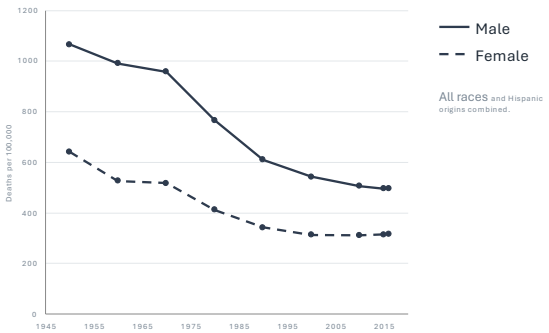
Has the mortality rate decreased in the U.S. ?




Case, A. and Deaton, A. 2020. Deaths of Despair and the Future of Capitalism

**Mortality rate among people aged 45–54, 1950–2016**

All-cause mortality by sex, all races combined

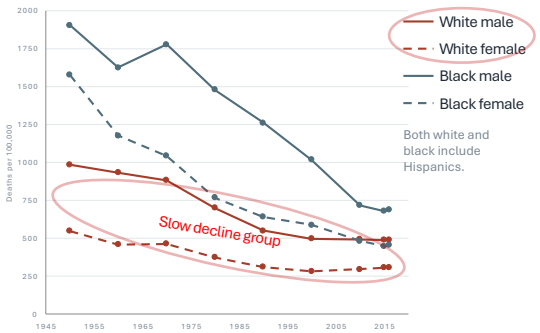



Source: NCHS, Health, United States 2017, Table 21 (death certificates). Published years only: 1950, 1960, 1970, 1980, 1990, 2000, 2010, 2015, 2016.

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**Mortality rate among people aged 45–54, 1950–2016**

All-cause mortality by sex, white and black



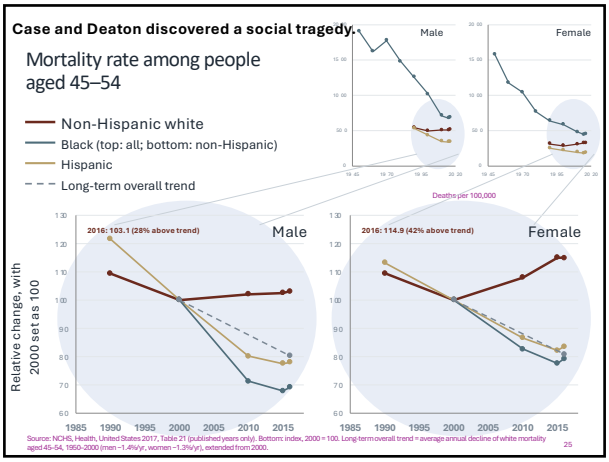

Source: NCHS, Health, United States 2017, Table 21 (death certificates). Published years only: 1950, 1960, 1970, 1980, 1990, 2000, 2010, 2015, 2016.

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**Case and Deaton discovered a social tragedy.**

Mortality rate among people aged 45–54

- Non-Hispanic white
- Black (top: all; bottom: non-Hispanic)
- Hispanic
- Long-term overall trend




Source: NCHS, Health, United States 2017, Table 21 (published years only). Bottom: index, 2000 = 100. Long-term overall trend = average annual decline of white mortality aged 45–54, 1950–2000 (men −1.4%/yr, women −1.3%/yr), extended from 2000.

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## Annual rate of change in mortality, aged 45–54

| Annual change | 1950–70 | 1970–90 | 1990–2000 | 2000–10 | 2010–16 |
|---|---|---|---|---|---|
| White men * | −0.5% | −2.4% | −0.9% | +0.2% | +0.2% |
| White women * | −0.8% | −2.0% | −0.9% | +0.8% | +1.0% |
| Hispanic men | — | — | −2.0% | −2.2% | −0.5% |
| Hispanic women | — | — | −1.2% | −1.4% | −0.6% |
| Black men | −0.3% | −1.7% | −2.1% | −3.5% | −0.7% |
| Black women | −2.1% | −2.5% | −0.8% | −2.0% | −0.9% |

- 1970–90: all groups declined by 1.7–2.5% per year
- 1990s: the decline among whites slowed to about 1% per year (yellow column)
- After 2000: only whites stalled or rose (red), most clearly among women; other groups kept declining

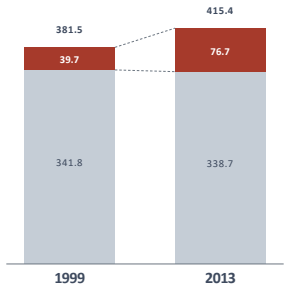
Source: calculated from NCHS, Health, United States 2017, Table 21. Annual change = log difference ÷ years. * White = all whites (incl. Hispanics) through 1990, non-Hispanic whites from 1990. Black includes Hispanics throughout.

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## Causes of Death

**Three causes doubled; all others held flat**

Death rate per 100,000, non-Hispanic whites aged 45–54 (US)

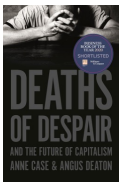



**Poisoning, suicide, liver disease**
**39.7 → 76.7**
nearly doubled (+37.0)

**All other causes**
**341.8 → 338.7**
slightly down (−3.1)

Source: Case & Deaton (2015), PNAS, Table 1. 1999 rates = 2013 rate − 1999→2013 change. "All other" = all-cause minus the three causes.

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Case, A. and Deaton, A. 2020. Deaths of Despair and the Future of Capitalism

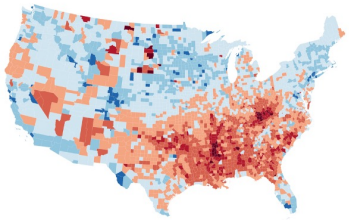
Case and Deaton compiled their findings into a book and published them under the title *Death by Despair.



## Topic 2

Spatial Distribution of a Demographic Parameter

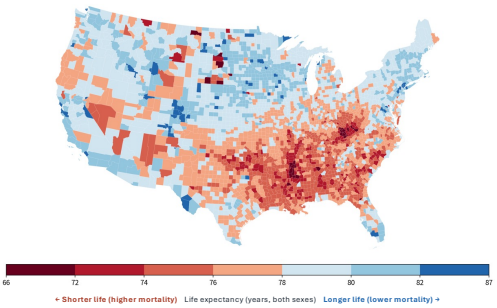




https://americaninequality.substack.com/p/life-expectancy-and-

## Life expectancy by county, United States (2014)

**Step 1: Distribution**




Dark red counties form a belt along the lower Mississippi, Appalachia (eastern Kentucky, West Virginia) and the Southeast, with scattered spots on Native American reservations in the Dakotas.

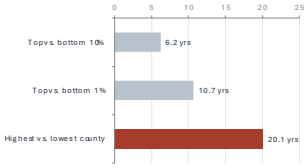
Source: IHME, United States Life Expectancy and Age-specific Mortality Risk by County 1980–2014 (Dwyer-Lindgren et al. 2017), 2014 values obtained via JsonOfCounties (GitHub). County boundaries: plotly/datasets (US Census Bureau). Alaska and Hawaii omitted. Grey = no data.

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## Geographic gaps in life expectancy

**Step 1: Size of gap**

**Gaps in county life expectancy (2014)**




**National average: 79.1 years**

- Highest: Summit County, Colorado, about 87
- Lowest: Oglala Lakota County, South Dakota, 66.8 (reservation)
- **Low-LE counties form a belt across the South and Appalachia**
- All 13 counties where LE fell in 1980–2014 are in the South (8 in Kentucky)

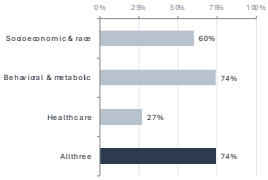
Source: Dwyer-Lindgren et al. (2017) JAMA Intern Med 177:1003–11 (3,110 counties, 1980–2014); American Inequality (Ney 2021). For the county map, see Figure 1 of the paper or the IHME website.

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**Step 2: Factors**

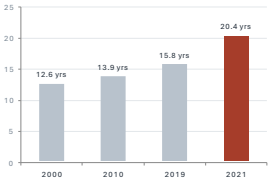
This reflects a social problem.

**Share of county variation in LE explained**



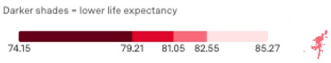

And it is growing year by year.

**"Ten Americas": largest group gap**




Source: Dwyer-Lindgren et al. (2017); Dwyer-Lindgren et al. (2024) Ten Americas, Lancet 404:2299–313. Behavioral/metabolic risks alone and all three groups combined both explain 74% because the factors overlap strongly.

32



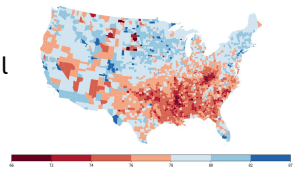

This is happening in **developed countries** that are failing to manage their societies effectively.



https://www.health.org.uk/resources-and-toolkits/data-tools/evidence-hub/map-of-life-expectancy-at-birth-by-constituency

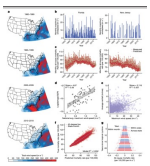
In fact, in the case of US, there was another hidden factor linked to human mortality.

That factor was mesoscale geographical conditions.



**Mortality caused by tropical cyclones in the United States**

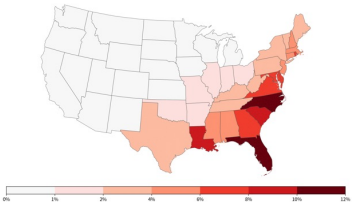
Rachel Young & Solomon Hsiang



*Nature* **volume 635**, pages121–128 (2024)

Hurricane-related disasters, linked to social issues, were causing an increase in mortality rates in certain regions.

**Cyclone-attributable deaths by state, 1950–2015**




Cyclone-attributable excess deaths as a share of all deaths (estimate)

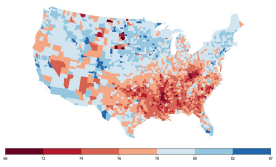
**Top: Florida 10.8%, North Carolina 10.5%, South Carolina 9.3%, Louisiana 8.8%.** Fades inland from the Atlantic and Gulf coasts; 0% in the West, which has no cyclone exposure.

Simplified recalculation: the linear impulse response in Fig. 2a (172 months) applied to state-by-month wind, population and deaths from the replication data of Young & Hsiang (2024) (Zenodo 10.5281/zenodo.11662454, CC BY 4.0). Broadly consistent with Fig. 4a (FL 13%, NC 11%, SC 9%, LA 8%).

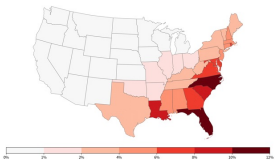
35

This hidden cause of death was hidden beneath the overall mortality pattern.

**Life expectancy by county (2014)**



**Cyclone share of deaths (by state)**

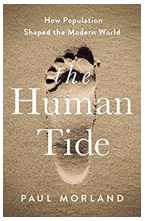


Source: left = IHME (Dwyer-Lindgren et al. 2017); right = simplified recalculation from Young & Hsiang (2024) data. Left is by county, right by state; spatial resolution differs.

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Topic 3

Demographical Insight on International Relations



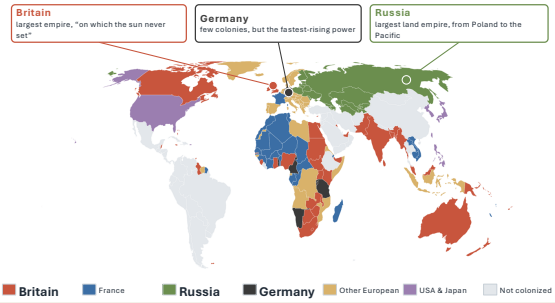

In **chapter 4**, "The German and Russian Challenges," it is discussed that **Germany** and **Russia** sought to catch up with **Britain**'s industrialization in 19th-century Europe.

The **industrialization** and **militarization** of these nations were **linked to their respective population structures** and **growth rates**, which in turn influenced the political balance in Europe.

Morand, P. 2019. Human Tide

## 1914: a few European powers ruled the world

Europeans and their descendants controlled **84% of the globe.**




Asia too: British India · Dutch East Indies · French Indochina · Russian Central Asia · concessions in China

***How did these few rivals come to rule — and why did they fight?***

84%: P. T. Hoffman (2015) Why Did Europe Conquer the World? Princeton UP (share of the globe under European or European-descended control by 1914). Map: approximate 1914 control shown on modern borders (Natural Earth); simplified.

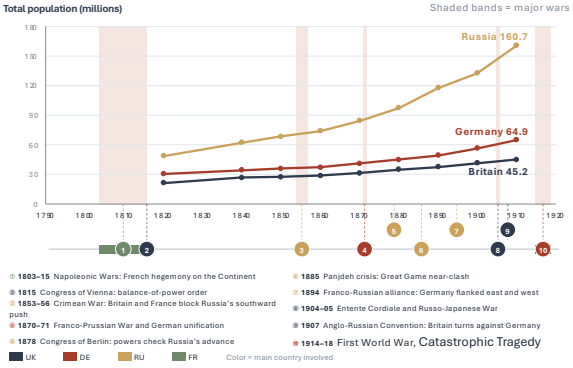
## Grand strategies of four powers

| Britain | Russia | Germany | France |
|---|---|---|---|
| Core principle<br>**Prevent a continental hegemon; control the seas, trade routes (and India)** | Core principle<br>**A plains state without natural frontiers seeks security through expansion and buffer zones** | Core principle<br>**In the middle of the Continent (Mittellage): avoid encirclement and a two-front war** | Core principle<br>**Facing a more populous rival (Germany), rely on allies at its back and on colonies** |
| • Balance of power: side against the strongest continental state (offshore balancing)<br>• Naval supremacy: two-power standard (1889); guarantee of Belgian neutrality (1839)<br>• Mackinder (1904): if railways bind the Eurasian "Heartland", land power will threaten sea power | • South: the Black Sea, the Straits and warm-water ports (Russo-Turkish Wars, Crimean War)<br>• Central Asia: securing the frontier against nomads and pressuring British India (conquest of the khanates)<br>• East: Amur and Maritime provinces (1858/60), Trans-Siberian Railway, Manchuria | • Bismarck: isolate France, ally with Austria and Russia (Three Emperors' League, Reinsurance Treaty); play the "satiated power" to prevent encirclement<br>• Wilhelm II: lets the Reinsurance Treaty lapse (1890) → Franco-Russian alliance (1894); Weltpolitik and the navy turn Britain hostile too<br>• Schlieffen Plan: if encircled, defeat one enemy quickly first | • After 1815: regain great-power status within the Vienna system<br>• After 1871: recover Alsace-Lorraine and secure against Germany; place allies behind Germany (Franco-Russian alliance 1894, Entente Cordiale 1904)<br>• Offset population stagnation with empire and capital (loans to Russia) |
| Across the 19th–20th centuries<br>**The target shifts (France → Russia → Germany → USSR), but the principle stays the same** | Across the 19th–20th centuries<br>**Defensive expansion, tsarist or Soviet: hold the periphery to protect the core** | Across the 19th–20th centuries<br>**Opposite means (diplomacy vs. speed), same problem: the two-front war** | Across the 19th–20th centuries<br>**"Allies behind the enemy" continues to the interwar Little Entente** |

Summary based on H. J. Mackinder, "The Geographical Pivot of History", Geographical Journal 23 (1904), and standard accounts.

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## Population trends and major events leading to conflict



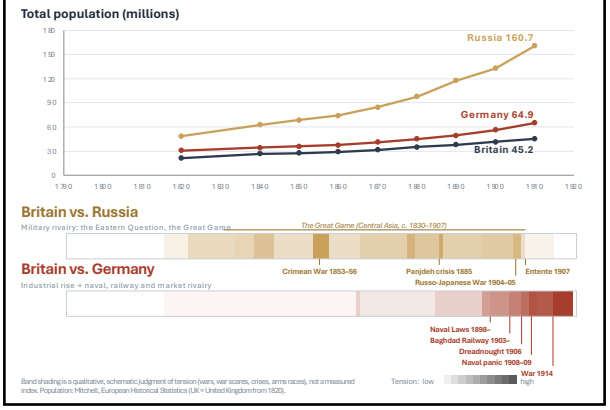

① **1803–15** Napoleonic Wars: French hegemony on the Continent
② **1815** Congress of Vienna: balance-of-power order
③ **1853–56** Crimean War: Britain and France block Russia's southward push
④ **1870–71** Franco-Prussian War and German unification
⑤ **1878** Congress of Berlin: powers check Russia's advance
⑥ **1885** Panjdeh crisis: Great Game near-clash
⑦ **1894** Franco-Russian alliance: Germany flanked east and west
⑧ **1904–05** Entente Cordiale and Russo-Japanese War
⑨ **1907** Anglo-Russian Convention: Britain turns against Germany
⑩ **1914–18** First World War, Catastrophic Tragedy

UK DE RU FR Color = main country involved

Population: attached file (Mitchell et al.). UK = United Kingdom from 1820. For details of events, see the timeline on slide 2.

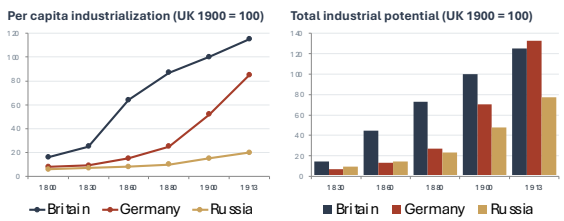
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## Population and Britain's two rivalries




Band shading is a qualitative, schematic judgment of tension (wars, war scares, crises, arms races), not a measured index. Population: Mitchell, European Historical Statistics (UK = United Kingdom from 1820).

## Population × industrialization = power




**Per capita industrialization (Bairoch):** Estimated manufacturing output (including handicrafts) divided by population, indexed to UK 1900 = 100. A gauge of how much manufactured output each person produces.

**Observations**

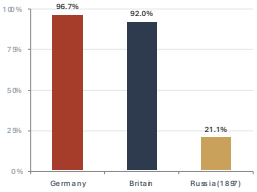
- Germany: per capita level rose 3.4-fold from 1880 to 1913; combined with population growth, its total overtook Britain's by 1913 (1.06×).
- Russia: largest population, but per capita level one sixth of Britain's; total only about 60% of Britain's.
- 1830–1860: only Britain grew, widening the gap with Germany and Russia (the material basis of the Pax Britannica).

Source: per capita = Bairoch (1982) Table 9 (reproduced in Baldwin & Martin 1999, NBER WP 6904). Total = per capita level × population (attached file; 1910 values used for 1913). Close to Kennedy (1987) totals (1913: UK 127, DE 138, RU 77).

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## Converting population into power

Literacy (late 19th–early 20th c.)




| | Britain | Germany | Russia |
|---|---|---|---|
| Primary education | Education Act 1870; compulsory from 1880 | Compulsory schooling in Prussia since the 18th century | Universal primary education legislation begun 1908 |
| Industrialization model | Private-led, free trade | Backed by state and banks; protective tariffs from 1879 (13% in 1913) | State-led, high tariffs (84% in 1913) |
| Population structure | Sends population abroad through emigration | Young population and urbanization | Over 80% rural, high fertility |

Britain "over 92%" (lower bound from a US Census Bureau comparison putting UK illiteracy below the US 7.7%)

**Inference**
To become national power, population needs literate workers and soldiers and the industry to employ them. Germany linked its well-educated population to industry; Russia's "conversion rate" was low for its size.

→ **Population is necessary but not sufficient for national power.**

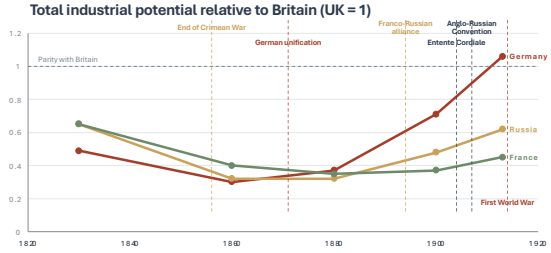
Source: Russia = 1897 census (literacy 21.1%); Germany = comparative figure cited in Topij (2011) Zapiski Historyczne 76(4); UK = US Bureau of the Census, Illiteracy in Foreign Countries; tariffs = Bairoch (1988) (NBER WP 6804 Table 8).



War depends on the mutual will and capabilities of the countries involved.

## Power balance and diplomatic turning points

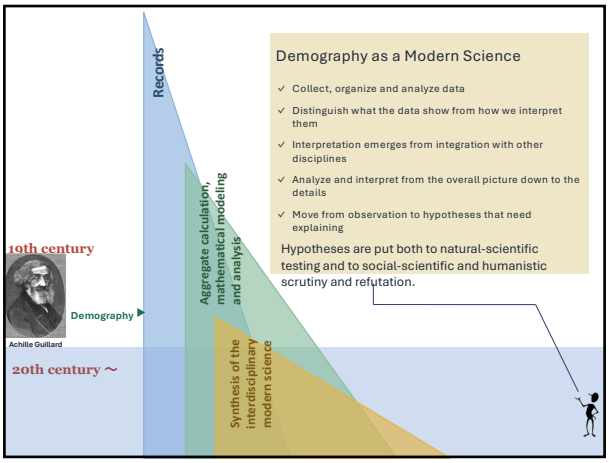
Total industrial potential relative to Britain (UK = 1)




- 1856–1880: Britain had about 3 times the power of Germany or Russia; its main concern was Russia (Eastern Question, Great Game).
- After 1900: Germany's ratio rose sharply, surpassing Britain in 1913. Britain settled with France (1904) and Russia (1907), lining up against Germany.

Ratios computed from Bairoch's per capita industrialization × population (slides 5 and 8). Timeline: slide 2.






### Demography as a Modern Science

- ✓ Collect, organize and analyze data
- ✓ Distinguish what the data show from how we interpret them
- ✓ Interpretation emerges from integration with other disciplines
- ✓ Analyze and interpret from the overall picture down to the details
- ✓ Move from observation to hypotheses that need explaining

Hypotheses are put both to natural-scientific testing and to social-scientific and humanistic scrutiny and refutation.

## Homework

- ➢ Work through the "Materials for Homework" on the "Introduction Website".
- ➢ Please write an essay report on a topic related to the materials.

Essay report topic

Will humankind eventually disappear?

**Deadline:** December 1, 17:00

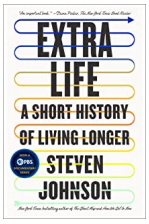
Submission: UTOL UTokyo LMS

File format: PDF

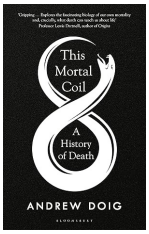
Welcome to add your opinions and impressions on my class.

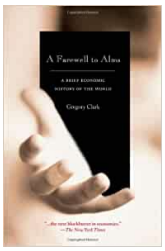
The amount you write is up to you.

Some recommended books to study




Epidemiology, Public Health, The History of Hunger and Changes in Life Expectancy

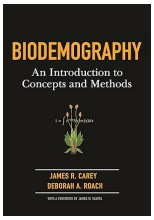



"A farewell to Alms"
Clark, G. 2008

A Comprehensive Review of Demography, Economics, and Sociology
A 100,000-Year History of the World Economy

Some recommended books to study Human-Demography and Theoretical Population Ecology







終わり